Class 19: Hedge Funds II

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Today's Question

Your family office is considering a fund-of-funds. What’s the extra layer of fees buying you?

Positioning

Where We've Been	Where We Are	Where We're Going
Hedge fund strategies	Evaluating hedge fund performance; due diligence	Yale case

Learning Objectives

By the end of today's class, you should be able to:

1. Explain challenges in measuring hedge fund α (survivorship bias, backfill, smoothing)
2. Describe the due diligence process for hedge fund investing
3. Articulate the case for and against hedge fund allocation in a portfolio

Recall our CAPM-based single-factor model:

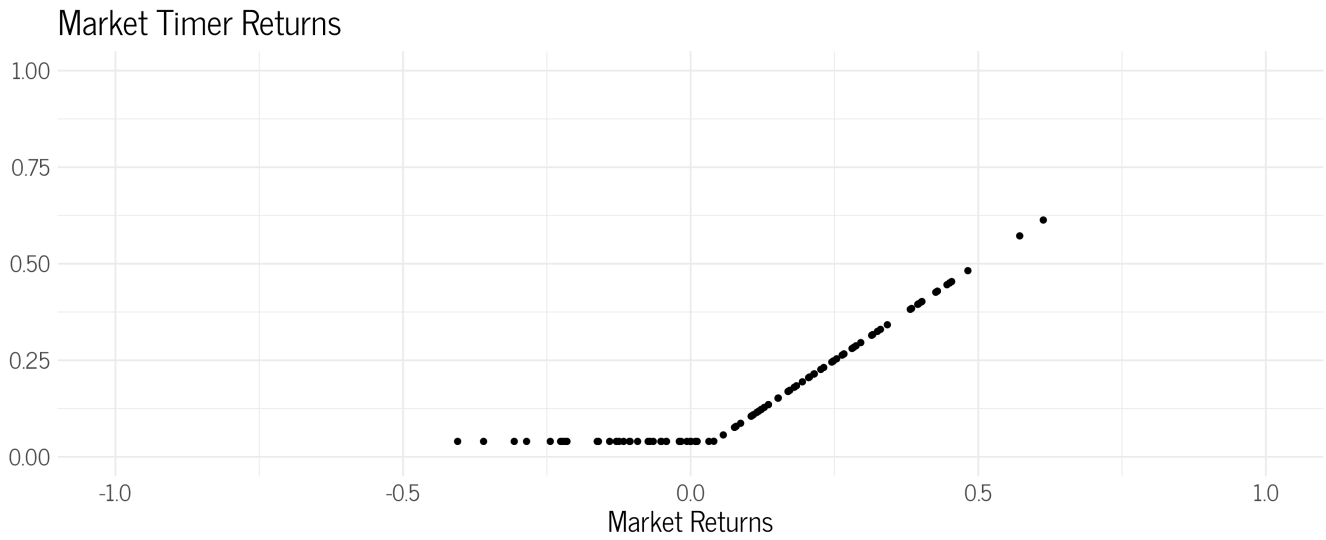
$$r_{i,t} - r_f = \alpha_i + \beta_i \left(r_{m,t} - r_f \right) + e_{i,t}$$

- For hedge funds, factor loadings are often **dynamic**
  - Consider market loading for a long-only market timer
  - If successful, β is only positive when market returns are high
  - This creates non-linear factor loadings

Dynamic Performance Evaluation

Suppose with perfect foresight, we can:

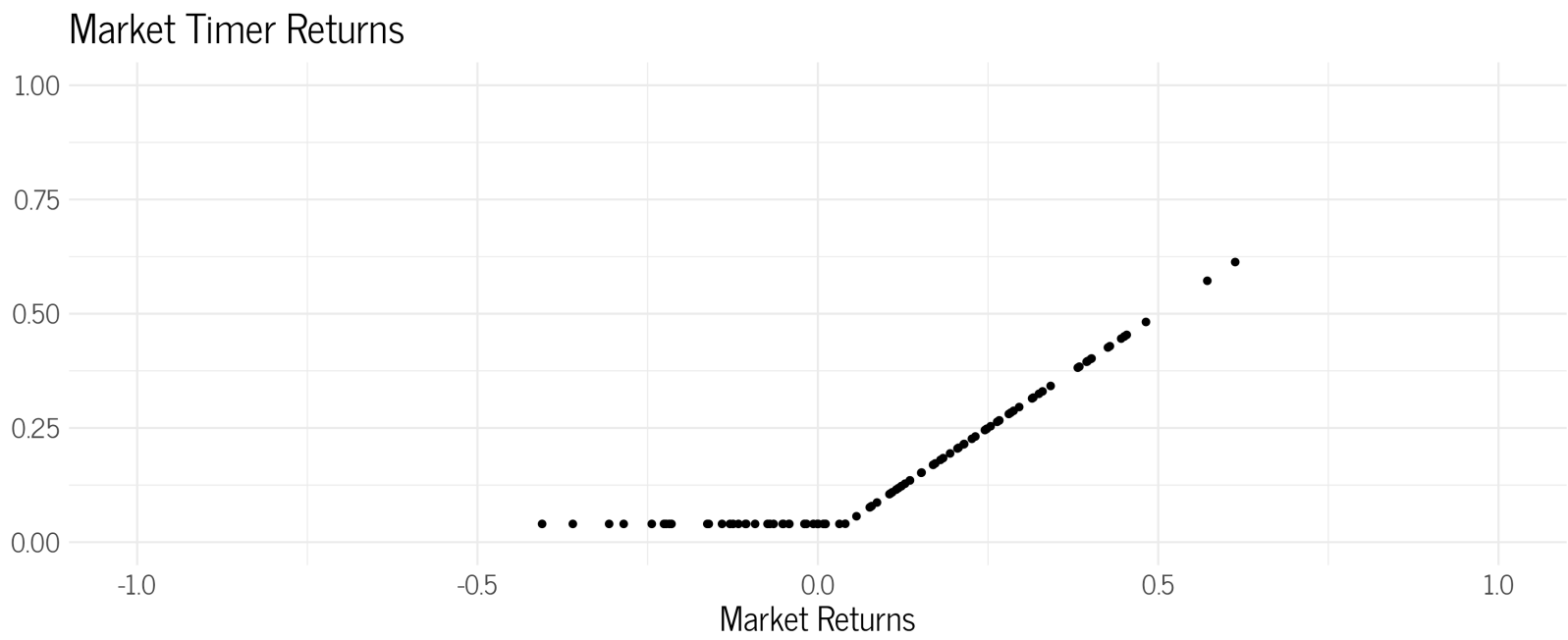
1. Buy the market when returns are above the risk-free rate of 4%
2. Hold T-bills when market returns fall below r_f



Dynamic Performance Evaluation

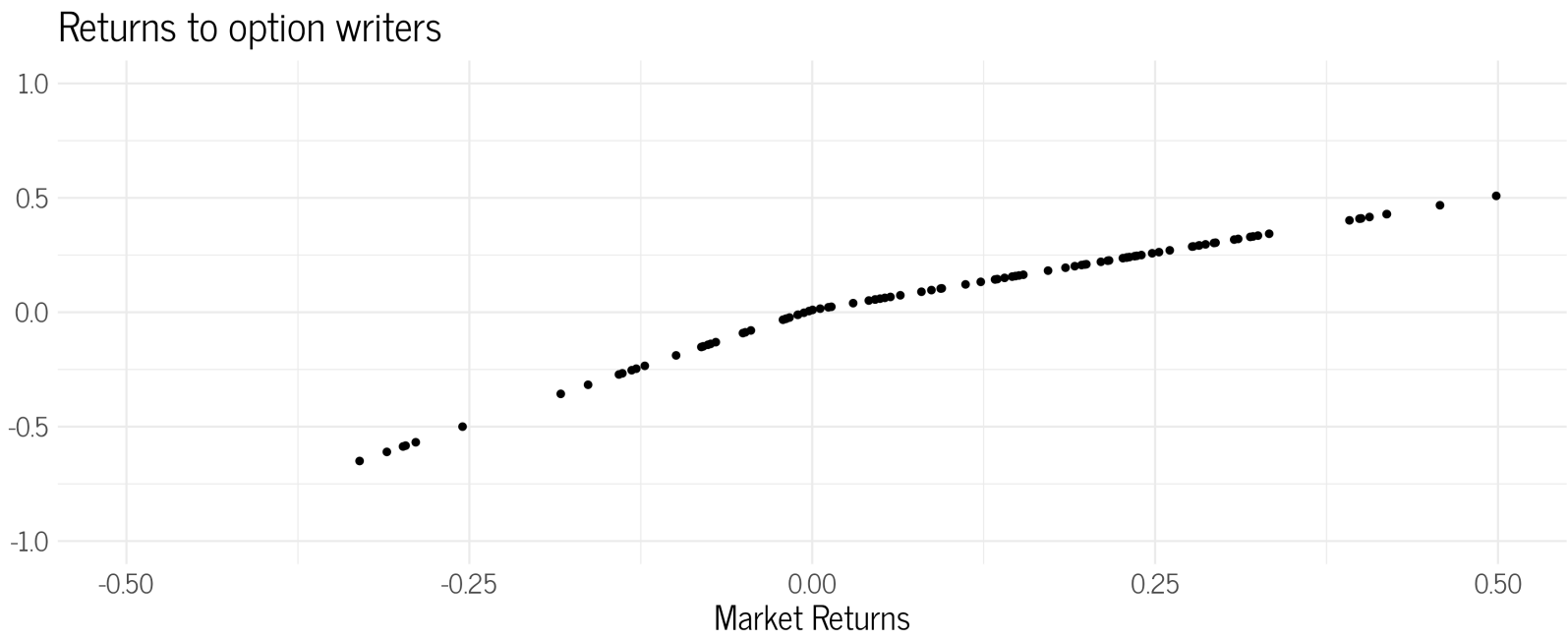
$$r_p = \alpha + r_f + \beta_1 (r_m - r_f) + \beta_2 (r_m - r_f)^2 + \varepsilon_p$$

- $\beta_2 > 0$ implies market-timing ability



Dynamic Performance Evaluation

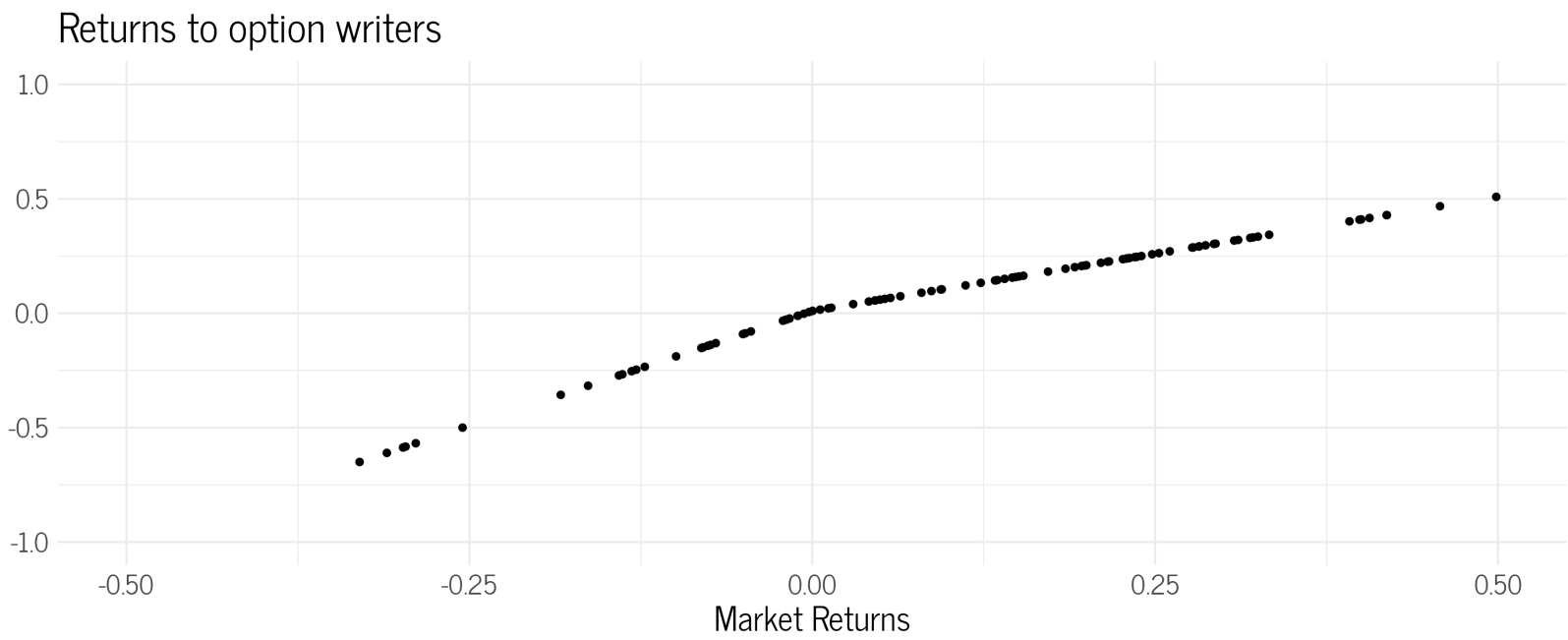
Alternatively, hedge funds may write out-of-the-money put options that exacerbate downturns.



Dynamic Performance Evaluation

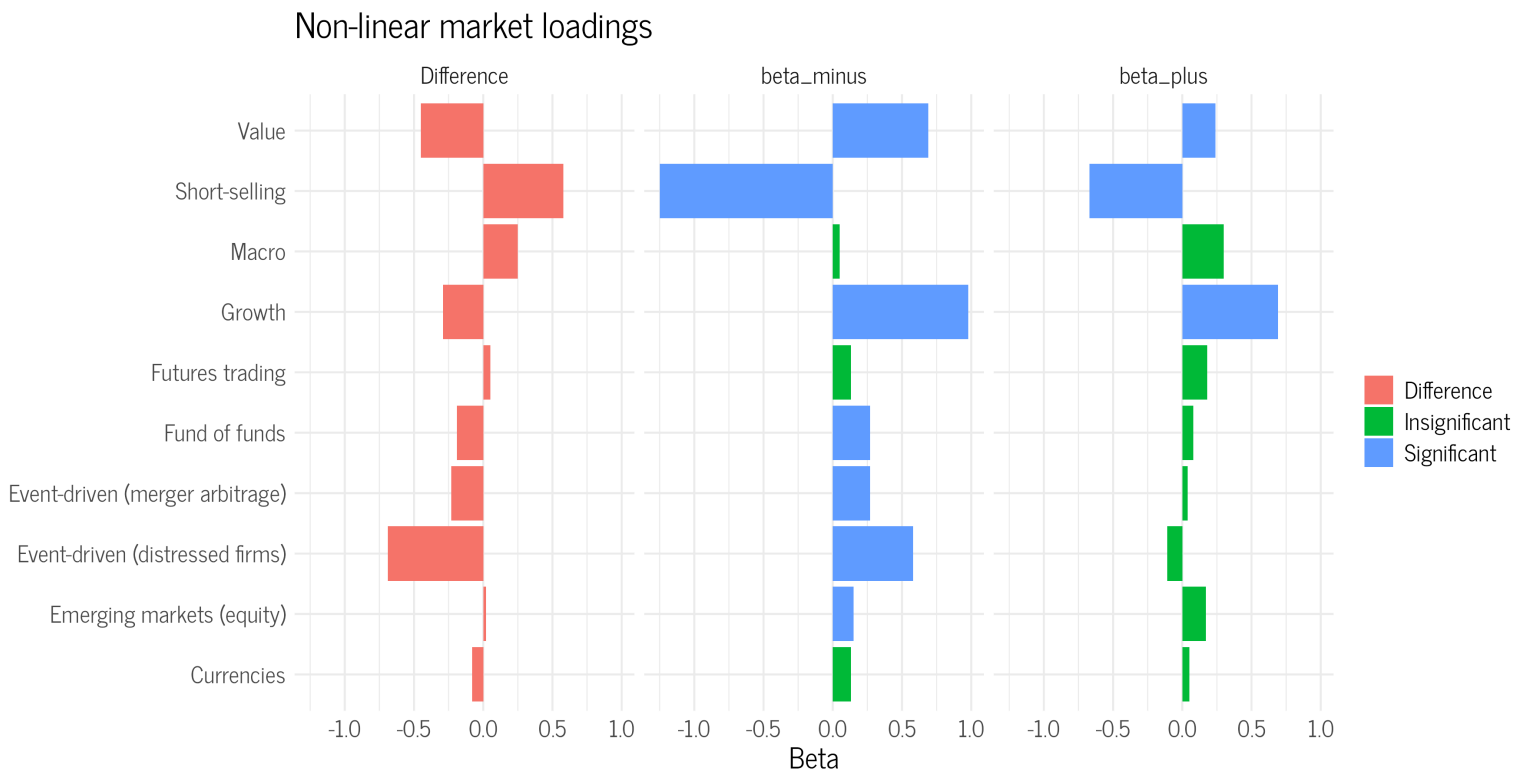
$$r_p = \alpha + r_f + \beta_1 (r_m - r_f) + \beta_2 (r_m - r_f)^2 + \varepsilon_p$$

- $\beta_2 < 0$ may imply option writing



Non-linear Loadings Across Asset Classes

$$\tilde{r}^e_{p,t} = \alpha_p + \beta^+ \max(\tilde{r}^e_{m,t}, 0) + \beta^- \min(\tilde{r}^e_{m,t}, 0) + \varepsilon_{p,t}$$



Why Hedge Fund Alpha Is Hard to Measure

A reported α can be misleading for **several different reasons**:

- **Dynamic betas:** a linear model can miss timing or option-like payoffs
- **Survivorship bias:** failed funds disappear, making the average survivor look too good
- **Backfill bias:** managers often start reporting only after a strong early track record
- **Smoothing:** illiquid assets or discretionary marks can make returns look artificially stable

Bottom line: "high alpha" may reflect model misspecification, selection bias, or stale pricing — not just skill.

Smoothing and Illiquidity

Hedge fund returns can look **too smooth** when positions are hard to value or are marked infrequently.

- Reported volatility looks lower than true economic volatility
- Reported correlations and betas can look artificially low
- Sharpe ratios can therefore look too high
- Losses may show up **late**, often when liquidity is worst

Investor implication: smooth returns are not always a sign of low risk; sometimes they are a sign that risk is hard to see.

Due Diligence: What Are You Actually Checking?

Good hedge fund due diligence is both **investment** and **operational**.

1. **Strategy & edge:** What is the source of returns? Why should it persist?
2. **People & process:** Who makes decisions? How repeatable is the process?
3. **Risk & liquidity:** What happens in bad states? How levered is the strategy? Can investors redeem?
4. **Valuation & operations:** Who prices positions? Who is the administrator, auditor, and custodian?
5. **Incentives & governance:** Fee terms, high-water marks, co-investment, transparency, and controls

A common mistake is to focus only on past returns rather than on whether the organization can survive stress.

Fund-of-Funds: What Are Fees Buying?

Potential Benefits	Potential Costs
Manager selection, access to otherwise closed funds, diversification across strategies	A second layer of management and incentive fees
Professional monitoring, due diligence, operational vetting	More distance from underlying positions and less transparency
Portfolio construction across managers and styles	Liquidity constraints may compound if underlying funds are gated
Useful for investors without internal alternatives staff	Net returns can disappoint if the value added is mostly commoditized

Answer to today's question: ideally, the extra fees buy better selection, diversification, and oversight — but they are only worth it if those services add more value than they cost.

Key Takeaways

- Hedge fund performance is hard to evaluate because exposures are often **non-linear** and **time-varying**
- Database biases and smoothing can make reported performance look better than true economic performance
- Due diligence therefore has to go beyond return histories and ask how the strategy, organization, and controls actually work
- Hedge funds can add value in a portfolio, but only if investors are realistic about fees, liquidity, and what is actually being purchased

Next Class: April 14

- Topics: Yale Endowment Case
  - Endowment investment philosophy
  - Asset allocation decisions
  - Alternative investments

- Matt Levine reading: TBD